

Eric R. Xu, Ph.D.

SENIOR DATA PROFESSIONAL | ON-PREM & CLOUD DATA PLATFORMS

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PROFESSIONAL SUMMARY

Senior data engineer with 10+ years building and supporting enterprise-class data pipelines across on-premises data warehouses and cloud data lakes at Apple, Meta, DoorDash, and Inmar. Advanced Python and SQL across ETL/ELT workflows, including LLM/RAG applications over relational and non-relational databases. Operates independently in lightly-guided environments, owning enterprise data solutions end to end and presenting technical direction to stakeholders and management.

TECHNICAL SKILLS

Data Engineering & Warehousing: Python · SQL · ETL/ELT & package development · Airflow · Spark · Presto/Trino · Kubernetes · dimensional modeling · enterprise data warehouse · BI/DW

Cloud & Architecture: AWS S3 · DigitalOcean · Apache Iceberg · Hive · PostgreSQL · MySQL · Elasticsearch · Medallion architecture · webhook-driven integrations

Quality, BI & AI: monitoring/alerting · anomaly detection · data-quality validation · Metabase · Tableau · Looker · LLM/RAG workflows · Slack API / Block Kit

EXPERIENCE

TileSense — Riichi Mahjong Trainer (Independent Project)

2026 – Present

- Built and shipped a cross-platform Flutter app with a PostgreSQL backend on DigitalOcean, instrumented with in-app eventing to capture gameplay and spaced-repetition learning signals.
- Built player-progress dashboards in Metabase over the app's Postgres event tables, tracking drill accuracy, review cadence, and retention across sessions.

Apple — Data Engineer (contracted by TEKsystems)

03/2025 – Present

- Build and support production Python and SQL (Presto/Trino) ETL, ingestion, and monitoring datasets spanning on-prem and cloud sources, improving real-issue detection by 35% year over year while reducing false positives by 30–40%.
- Developed webhook-driven Slack alert-enrichment and RAG-based triage services on Kubernetes, rendering incoming alerts as interactive Block Kit investigation context and increasing alert engagement by 20% at stable volume.
- Partner with Engineering and Product stakeholders to translate ambiguous requirements into metric definitions, scalable pipelines, Tableau reporting, and clear technical recommendations.

DoorDash — Analytics Data Engineer (contracted by TEKsystems)

08/2024 – 02/2025

- Designed layered analytics transformations, serving datasets, and self-service BI for in-store (brick-and-mortar hospitality and retail) deals, supporting market expansion, redemption, conversion, and cohort analysis.
- Defined reusable metric logic, automated data-quality checks, and experiment launch monitoring across a Medallion-style raw-to-serving flow; informed UI strategies that improved conversion efficiency by 5–10%.
- Built forecasting and milestone-tracking reporting across 50K+ cumulative redemptions, surfacing deviations early and reducing recurring ad-hoc analysis.

Meta — Senior Analytics Data Engineer

07/2020 – 04/2024

- Designed and owned orchestrated Spark/Hive/Presto ELT pipelines and dimensional (star-schema) tables in a layered raw → curated → serving architecture supporting analytics, experimentation, and high-traffic consumer products.
- Built reusable reporting datasets and KPI frameworks, including a CPU-efficiency metric that drove a 15% increase in efficiency-focused code changes and reduced compute costs.
- Migrated legacy warehouse tables to Apache Iceberg and partnered with Data Scientists and Product teams to validate metrics, troubleshoot distributed pipeline failures, and restore trusted reporting.

Inmar Intelligence | OwnerIQ — Senior Data Engineer / Senior Data Scientist

07/2015 – 06/2020

- Led end-to-end delivery of 8+ production data and BI/DW products and ETL/ELT pipelines, from stakeholder requirements and technical design through validation, documentation, delivery, and adoption.
- Built large-scale transformation, recommendation, and targeting pipelines with Python, Airflow, Spark, Hive, Presto, AWS S3, MySQL, and Elasticsearch k-NN to generate commerce insights and ROAS gains.
- Implemented automated data-quality checks, recurring KPI reporting in Looker, and internal analytics tools that standardized commerce measurement and reduced manual analysis.

EDUCATION

Certificate, Data Science & Machine Learning — MIT Schwarzman College of Computing, Nov 2024

Ph.D., Psychological & Brain Sciences — Dartmouth College

B.S., Biomedical Engineering (Biomechanics) — Case Western Reserve University